is the underlying principle behind the term "preference": an agent demonstrates a strong tendency to cooperate with a particular endpoint agent to maximize their utility.

To deeply investigate the learning process, we further introduce the *sub-preference graphs* based on P-GFGs, which aim to reformulate previous learning states and analyze the learning behavior of the algorithm. Suppose that there is a set of sequentially generated strategies $\mathcal{N}_n = \{1, 2, \cdots, n\}$, where the index also represents the number of iterations for simplicity. For each previous iteration $i < n$, the sub-preference game form graph is denoted as $\{\mathcal{N}_i, \mathbf{E}'_i, \mathbf{w}_i\}$, where $\mathcal{N}_i = \{1, 2, \cdots, i\}$ is the set of strategies in iteration i, and $\mathbf{E}'_i, and\ \mathbf{w}_i$ are the corresponding edges and weights.

The semantics of the preference graph is that a strategy or node i prefers to play with the tailed node to achieve the highest results. In other words, the more in-edges one node has, the more cooperative ability this node can achieve. Ideally, if one strategy can adapt well to all others, all the other strategies in the preference graph will point to this strategy. To evaluate the adaptive ability of each node, the centrality concept is introduced into the preference graph to evaluate how a node is preferred.

Definition 4.2 (Preference Centrality)**.** Given a P-GFG $\{\mathcal{N}, E', \mathbf{w}\}$, preference centrality of $i \in \mathcal{N}$ is defined as,

$$\eta(i) = 1 - \text{norm}(d_i),$$

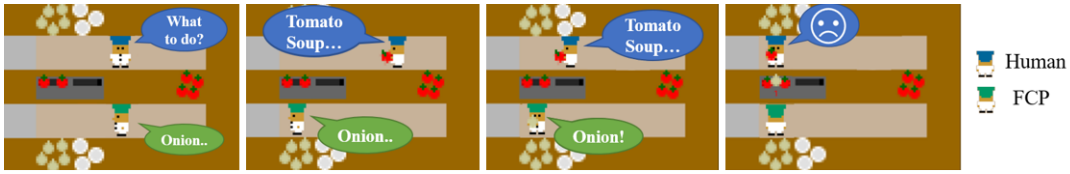
where d_i is a graph centrality metric to evaluate how the node is preferred, and norm $:= \mathbb{R} \to [0, 1]$ is a normalization function.

Note that the d is a kind of centrality that could evaluate how much a node is preferred. A typical example of d is the centrality of degrees, which calculates how many edges point to the node. In this work, our primary choice for the graph centrality metric is in-degree centrality.

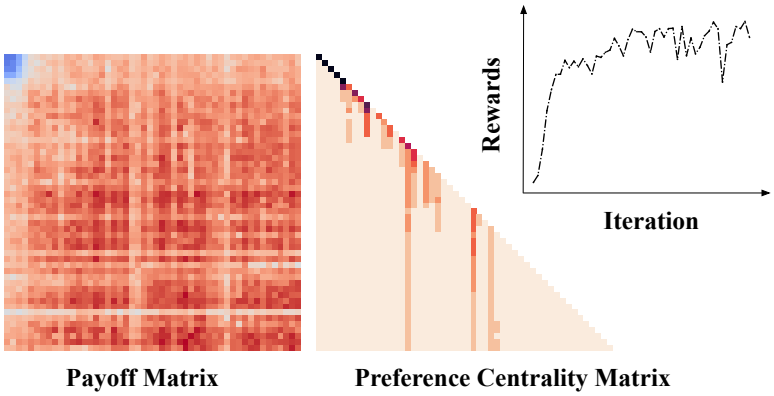
Fig. 1 provides examples of the learning processes of two algorithms, one with and one without a cooperative incompatibility issue. In Fig. 1(a), illustrative of the algorithm possessing this cooperative incompatibility, the updated strategy is observed to fall short in elevating the cooperative utility achieved with other strategies after the second generation. There are no edges pointing towards the node which validates this further, and is confirmed by the preference centrality matrix, where the strategy at hand is marked with a $\eta = 1$. This suggests that no nodes wish to collaborate with the updated strategies. On the other hand, Fig. 1(b), devoid of cooperative incompatibility, portrays a different scenario. In this depiction, every other strategy within the preference graph directs towards the most recent strategy, signifying the continuous enhancement of the cooperative capability.

4.2 Motivating Examples

The challenge of training AI agents to effectively coordinate with unfamiliar AI counterparts or human players constitutes a substantial impediment. Predominantly trained through reward maximization methods, AI agents typically demonstrate deterministic strategies which often diverge from those employed by human players or other AI agents, thereby resulting in a coordination failure with previously unseen players (Ye et al., 2020; Gao et al., 2023; Yu et al., 2023). Fig. 2(a), proposed by HSP (Yu et al., 2023), illustrates a motivating example in the Overcooked game of how an agent player could collaborate with some players and



(a) The illustration showcases an instance of an FCP agent's collaboration failure with specified human participants. This figure is referenced from HSP (Yu et al., 2023).



(b) Cooperative incompatibility issue in MEP training process.

Figure 2: Motivating examples. Fig. 2(a) features an analysis conducted by HSP (Yu et al., 2023). The FCP agent converges to a fixed pattern of exclusively preparing onion soup, thereby failing to establish coordination with a human participant who prefers making tomato soup. Fig. 2(b) shows a training process of the MEP algorithm with several comparative incompatibility problems. The payoff matrix of each strategy during training and the corresponding preference centrality matrix of the MEP algorithm in the Overcooked. A deeper shade of red in the payoff matrix signifies higher utility. The darker the color in the preference centrality matrix, the lower the centrality value, and the more other strategies prefer it.

fail to collaborate with other players. In this example of the Overcooked game, chefs could accomplish two recipes: one calls for three onions and the other demands three tomatoes. The FCP algorithm, however, tends to settle into a distinctive pattern of cooking solely onion soup (Yu et al., 2023). Consequently, this FCP agent can achieve high scores when paired with similarly styled human players. Nevertheless, its performance declines significantly when paired up with a player who prefers the tomato soup style, resulting in a disruption of the human player's tomato soup cooking plan, thereby failing to complete the required recipe. Further instances of miscoordination can be observed in real-world scenarios, such as autonomous driving, where the question may arise whether the policy should be aggressive or conservative.

Nevertheless, miscoordination issues exist in the training of some population-based ZSC algorithms. An analysis of the MEP algorithm (Zhao et al., 2021), as depicted in Fig. 2(b), reveals a cooperative incompatibility evident during the learning process in the Overcooked

environment (Carroll et al., 2020). The figure on the left represents the payoff matrix, where a deeper shade of red signifies higher utility. In the accompanying preference indegree centrality matrix, a darker color is indicative of a strategy being more preferred by others. Throughout the MEP learning process, despite a consistent improvement in mean rewards (as depicted in the upper-right of Fig. 2(b)), significant cooperative incompatibility problems arise post a certain training duration. At this juncture, many strategies show an inclination to engage with earlier strategies, represented by darker shades, over newer strategies, in an effort to secure higher rewards. Therefore, addressing this collaboration incompatibility is crucial to enhance the adaptability of AI agents in coordinating with unseen partners.

4.3 Cooperative Incompatibility

Fig. 1 provides examples of cooperative incompatibility, specifically concerning human-AI coordination and ZSC algorithm learning. Consequently, this section will delve deeper into a conceptual understanding of cooperative incompatibility and formally delineate this issue within the learning process, utilizing the tools previously introduced such as P-GFGs and preference centrality.

Definition 4.3 (Cooperative Incompatibility)**.** Cooperative incompatibility pertains to the occurrence wherein a participant, either an AI agent or a human player, is unable to align with certain specific partners, who may equally be AI agents or human players.

Cooperative Incompatibility in the learning of ZSC algorithms. A cooperative incompatibility issue arises in a population-based learning algorithm when the new strategy s_t, produced by the algorithm at step t, is unable to boost cooperative utility in conjunction with existing strategies in the population. This is to say, the preference centrality $\eta(t)$ of s_t is greater than 0. Ideally, an algorithm devoid of cooperative incompatibility issues will maintain a preference centrality of 0 at each step. This implies that the updated strategy enhances the collaborative utility of strategies within the population, prompting them to prefer cooperation with the updated strategy.

In this study, our aim is to resolve the cooperative incompatibility in the learning process of population-based algorithms, ultimately paving the way to tackle cooperative incompatibility issues in AI-AI and AI-human coordination.

4.4 Cooperative Open-Ended Learning Framework

To address the problem of cooperative incompatibility, we refine our comprehension of the objective by formulating empirical gamescapes (Balduzzi et al., 2019) from zero-sum games to our studied shared payoff GFGs. This provides us with a geometric representation of player strategies within a GFG $\{\mathcal{N}, \mathbf{E}, \mathbf{w}\}$, thereby capturing the diversity and adaptive capacity of cooperative strategic behaviour. However, it is not efficient to directly learn how to extend the EGS in common payoff games to cooperate effectively with unseen partners.

To conquer cooperative incompatibility, the natural idea is to learn with the mixture of cooperative incompatible strategies on the most recent population $\mathcal{N}$ to improve gamescape. Given a population $\mathcal{N}$, we present *cooperative incompatible solver* to assess how strategies collaborate, especially with those strategies that are difficult to collaborate with. The solver derives the cooperative incompatible distribution ϕ, where strategies that do not coordinate